

Project Title
Near-Duplicate Question Finder for Q&A Platforms

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The **Near-Duplicate Question Finder for Q&A Platforms** is a Data Structures and Algorithms (DSA) project designed to identify semantically similar or nearly identical questions efficiently in large-scale question-and-answer platforms. Duplicate questions often lead to redundant content, increased storage requirements, and reduced search efficiency. This project addresses the problem by combining efficient string processing, hashing techniques, and similarity analysis to detect near-duplicate questions while minimizing computational cost. The proposed system first preprocesses the input questions by normalizing the text and generating **k-word shingles (k-grams)** to capture local textual patterns. Each shingle is then converted into compact fingerprints using **double polynomial rolling hashing**, which reduces hash collisions and improves the reliability of similarity detection. To avoid expensive pairwise comparisons among all questions, **Locality-Sensitive Hashing (LSH)** is employed to group similar fingerprints into candidate buckets, significantly reducing the search space. Finally, an **exact common-substring analysis** is performed on the shortlisted candidate pairs to eliminate false positives and accurately verify duplicate questions.

The performance of the system is evaluated by comparing the **precision before and after the verification stage**, highlighting the improvement achieved through exact similarity verification. By integrating shingling, double hashing, LSH, and exact string matching, the proposed solution provides an efficient and scalable approach for duplicate question detection, making it suitable for online discussion forums, educational portals, knowledge-sharing websites, and customer support platforms.

Signature of the Guide